

Make This Noise Detector For A Noise-Free Zone

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Please Keep Silence! You must have seen this message written in such places as libraries and offices. But some people tend to, intentionally or unintentionally, make noise in such silence zones, disturbing others. What if there was a device that could help maintain silence in such silence areas?

```
//Silence_zone.ino
int senpin=A0;
int buzzer=8;
int light=13;
long val=0;
long average = 0;
int threshold=40;
String ans;
```

Fig. 1: Setting variables

```
void setup() {
  pinMode(senpin, INPUT);
  pinMode (light, OUTPUT);
  pinMode(buzzer, OUTPUT);
  Serial.begin(9600);
}
```

Fig. 2: Arduino code setting I/O pins

```
void loop() {
  if(Serial.available()!=0){
    ans=Serial.readStringUntil('\n');
    threshold=ans.toInt();
  }
  sensor();
}
```

Fig. 3: Setting loop function

```
void sensor(){
  for (int i=0; i < 160; i++) {
    average = average + analogRead(senpin);
  }
  val = average/160;
  average=0;
  delay (10);
  Serial.print("\t");
  Serial.print(val);
  Serial.print("\t");
  Serial.print("-----");
  Serial.println(threshold);
  if (val>= threshold){
    digitalWrite(light,HIGH);
    delay (2000);
    digitalWrite(light,LOW);
  }
}
```

Fig. 4: Arduino code sensor function

In this project, we make a sensitive smart noise detector that can identify loud talkers and alert us by displaying a red light and triggering a buzzer. You can also connect this device to your phone through a customised app to set sensitivity levels as per choice.

To start, the following components are needed:

- Arduino Uno
- Red LED
- Buzzer
- Sound sensor
- Some jumper wires
- 5V DC adaptor/5V power bank

Coding

Software program (silence_zone.ino) in Arduino handles all necessary func-

tions for noise detection. Set up the variables for storing values in Arduino code, as shown in Fig. 1.

Then, configure Bluetooth and I/O pin for input and output in setup() function. Here, Bluetooth baud rate is set to 9600, but you can try with 38400 default baud rate as well (Fig. 2).

Next, set loop functions for running the code repeatedly (Fig. 3). In this loop function, create an if() condition that listens to Bluetooth incoming string. Then, convert that string to integer value and set that integer value to threshold value.

In the next part of coding, create sensor() function to get sensor value (Fig. 4). For sensor value, take an average of several readings of detected noise levels. The formula for obtaining



Fig. 5: App layout

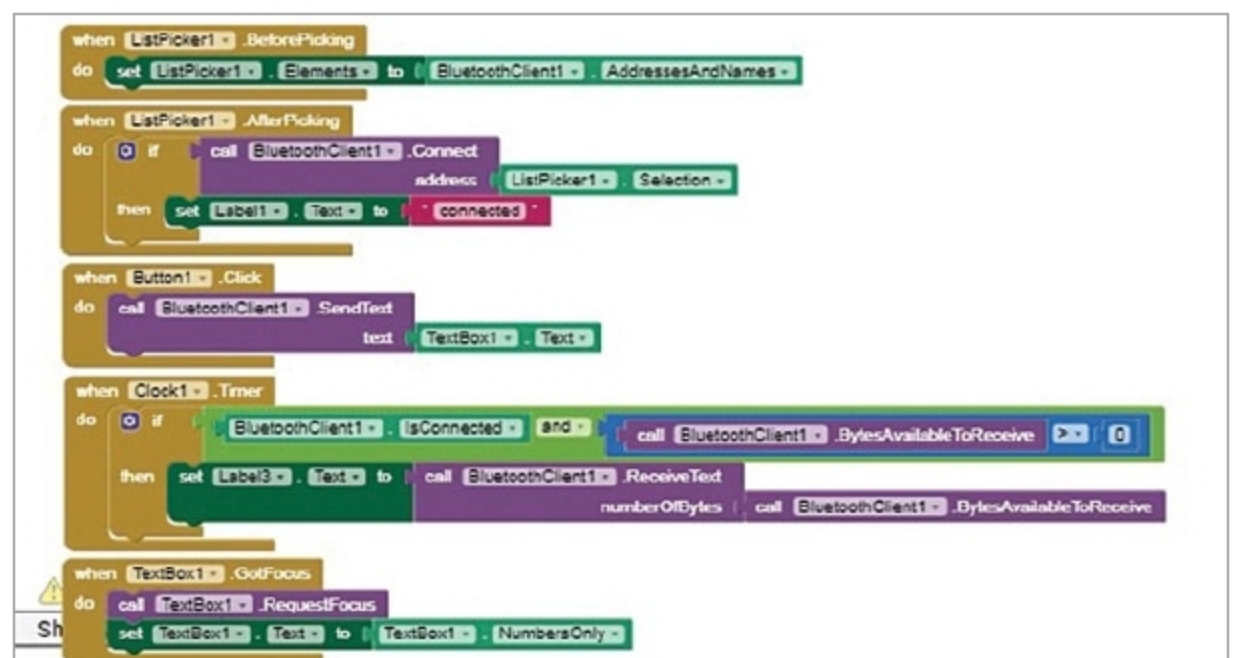


Fig. 6: MIT App Inventor code